

Ethyl acetate

Version	Revision Date:	SDS Number:	Date of last issue: 23.03.2022
1.1	28.11.2022	000000033638	Date of first issue: 23.03.2022

1. PRODUCT AND COMPANY IDENTIFICATION

Product name : Ethyl acetate

Product code : 000000000051011132

Manufacturer or supplier's details

Company : Celanese Pte Ltd

Address : 70 Anson Road #21-01/02 Hub Synergy Point
Singapore 079905

Telephone :

Emergency telephone number : CHEMTREC: +1 703 527 3887

Recommended use of the chemical and restrictions on use

Recommended use : Solvent

2. HAZARDS IDENTIFICATION**GHS Classification**

Flammable liquids : Category 2

Acute toxicity (Oral) : Category 5

Specific Target Organ Toxicity : Category 3 (Central nervous system)

GHS label elements

Hazard pictograms :



Signal word : Danger

Hazard statements : H225 Highly flammable liquid and vapour.
H303 May be harmful if swallowed.
H336 May cause drowsiness or dizziness.

Precautionary statements :

Prevention:

P210 Keep away from heat/ sparks/ open flames/ hot surfaces.
No smoking.

P261 Avoid breathing dust/ fume/ gas/ mist/ vapours/ spray.

P280 Wear protective gloves/ eye protection/ face protection.

Response:

P303 + P361 + P353 IF ON SKIN (or hair): Take off immediate-

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ly all contaminated clothing. Rinse skin with water/ shower.
P304 + P340 IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing.
P312 Call a POISON CENTER or doctor/ physician if you feel unwell.

Other hazards which do not result in classification

None known.

3. COMPOSITION/INFORMATION ON INGREDIENTS

Substance / Mixture : Substance

Components

Chemical name	CAS-No.	Concentration (% w/w)
ethyl acetate	141-78-6	>= 99.7

4. FIRST AID MEASURES

General advice : Do not leave the victim unattended.
Remove contaminated, soaked clothing immediately and dispose of safely
In any case show the physician the Safety Data Sheet

If inhaled : If breathed in, move person into fresh air.
If unconscious, place in recovery position and seek medical advice.
If symptoms persist, call a physician.

In case of skin contact : Wash off with soap and plenty of water.
Remove contaminated clothing and shoes.
Wash contaminated clothing before re-use.
If symptoms persist, call a physician.

In case of eye contact : Flush eyes with water as a precaution.
Remove contact lenses.
Protect unharmed eye.
Keep eye wide open while rinsing.
If eye irritation persists, consult a specialist.

If swallowed : Rinse mouth with water.
If swallowed, do not induce vomiting - seek medical advice.
Keep respiratory tract clear.
Do not give milk or alcoholic beverages.
Never give anything by mouth to an unconscious person.
If symptoms persist, call a physician.

Most important symptoms and effects, both acute and delayed : May cause drowsiness or dizziness.

Notes to physician : Treat symptomatically.

5. FIREFIGHTING MEASURES

Suitable extinguishing media : Foam
Dry chemical
Carbon dioxide (CO₂)

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Unsuitable extinguishing media	: High volume water jet
Specific hazards during fire-fighting	: Vapours are heavier than air and may spread along floors.
Hazardous combustion products	: Carbon oxides
Specific extinguishing methods	: Cool containers/tanks with water spray. Do not allow run-off from fire fighting to enter drains or water courses. Standard procedure for chemical fires. Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.
Special protective equipment for firefighters	: Wear self-contained breathing apparatus for firefighting if necessary.

6. ACCIDENTAL RELEASE MEASURES

Personal precautions, protective equipment and emergency procedures	: Keep away from heat and sources of ignition. Ensure adequate ventilation. Use personal protective equipment. Avoid inhalation, ingestion and contact with skin and eyes.
Environmental precautions	: Prevent product from entering drains. Prevent further leakage or spillage if safe to do so. If the product contaminates rivers and lakes or drains inform respective authorities.
Methods and materials for containment and cleaning up	: Soak up with inert absorbent material (e.g. sand, silica gel, acid binder, universal binder, sawdust). Keep in suitable, closed containers for disposal. Non-sparking tools should be used.

7. HANDLING AND STORAGE

Advice on protection against fire and explosion	: Normal measures for preventive fire protection. Take measures to prevent the build up of electrostatic charge. Use explosion-proof equipment.
Advice on safe handling	: For personal protection see section 8. Provide sufficient air exchange and/or exhaust in work rooms. Do not breathe vapours or spray mist. Avoid contact with the skin and the eyes. Smoking, eating and drinking should be prohibited in the application area. Provide adequate precautions, such as electrical grounding and bonding, or inert atmospheres. Ensure all equipment is electrically grounded before beginning transfer operations. Dispose of rinse water in accordance with local and national regulations.
Conditions for safe storage	: Keep tightly closed in a dry, cool and well-ventilated place. Store locked up.

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Electrical installations / working materials must comply with the technological safety standards.
 Handle and open container with care
 Further information on storage stability : No decomposition if stored and applied as directed.

8. EXPOSURE CONTROLS/PERSONAL PROTECTION

Components with workplace control parameters

Components	CAS-No.	Value type (Form of exposure)	Control parameters / Permissible concentration	Basis
ethyl acetate	141-78-6	PEL (long term)	400 ppm 1,440 mg/m ³	SG OEL
		TWA	400 ppm	ACGIH

Engineering measures : Local exhaust
 Use explosion-proof equipment.

Personal protective equipment

Respiratory protection : In case of mist, spray or aerosol exposure wear suitable personal respiratory protection and protective suit.
 Use NIOSH approved respiratory protection.

Filter type : Organic vapour type

Hand protection

Material : butyl-rubber

Break through time : 120 min

Glove thickness : 0.7 mm

Directive : Equipment should conform to EN 374

Protective index : Class 4

Remarks : The suitability for a specific workplace should be discussed with the producers of the protective gloves.

Eye protection : Safety goggles
 In addition to goggles, wear a face shield if there is a reasonable chance for splash to the face.
 Ensure that eyewash stations and safety showers are close to the workstation location.

Skin and body protection : Impervious clothing
 Choose body protection according to the amount and concentration of the dangerous substance at the work place.

Protective measures : Avoid contact with skin and eyes.
 Do not breathe vapours or spray mist.

Hygiene measures : When using do not eat or drink.
 When using do not smoke.
 Wash hands before breaks and at the end of workday.
 Handle in accordance with good industrial hygiene and safety practice.

9. PHYSICAL AND CHEMICAL PROPERTIES

Appearance : liquid

Colour : colourless

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Odour : fruity

Odour Threshold : 0.006 - 0.686 ppm

pH : not determined

Melting point/range : -83.8 °C

Boiling point/boiling range : 77.1 °C
(1,013 hPa)

Flash point : -4 °C
Method: closed cup

Upper explosion limit / Upper flammability limit : 11.5 %(V)

Lower explosion limit / Lower flammability limit : 2.2 %(V)

Vapour pressure : 379 hPa (50 °C)

Density : 0.9003 g/cm³ (20 °C)

Solubility(ies)

Water solubility : 80 g/l (25 °C)

Solubility in other solvents : completely soluble
Solvent: Acetone

miscible
Solvent: Benzene

miscible
Solvent: Diethylether

miscible
Solvent: Ethanol

Partition coefficient: n-octanol/water : log Pow: 0.68
measured data

Viscosity

Viscosity, dynamic : 0.451 mPa.s (20 °C)

Explosive properties : not applicable based on consideration of the structure

Surface tension : 24 mN/m, 20 °C

Molecular weight : 88.11 g/mol

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10. STABILITY AND REACTIVITY

Reactivity	:	No decomposition if stored and applied as directed.
Chemical stability	:	Stable under normal conditions.
Possibility of hazardous reactions	:	No decomposition if stored and applied as directed.
Conditions to avoid	:	Keep away from heat and sources of ignition.
Incompatible materials	:	Peroxides Strong acids Oxidizing agents Amines

11. TOXICOLOGICAL INFORMATION**Acute toxicity**

Not classified based on available information.

Components:**ethyl acetate:**

Acute oral toxicity	:	LD50 (Rabbit): 4,934 mg/kg Method: OECD Test Guideline 401
Acute inhalation toxicity	:	LC100 (Rat): > 22 ppm Exposure time: 6 h
Acute dermal toxicity	:	LD50 (Rabbit, male): > 20,000 mg/kg

Skin corrosion/irritation

Not classified based on available information.

Serious eye damage/eye irritation

Not classified based on available information.

Respiratory or skin sensitisation**Skin sensitisation**

Not classified based on available information.

Respiratory sensitisation

Not classified based on available information.

Components:**ethyl acetate:**

Species	:	Guinea pig
Method	:	OECD Test Guideline 406
Result	:	Does not cause skin sensitisation.

Germ cell mutagenicity

Not classified based on available information.

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Components:**ethyl acetate:**

Genotoxicity in vitro : Test Type: Ames test
Metabolic activation: with and without metabolic activation
Method: OECD Test Guideline 471
Result: negative

Test Type: In vitro cytogenicity study in mammalian cells
Test system: Chinese hamster cells
Metabolic activation: with and without metabolic activation
Method: OECD Test Guideline 473
Result: negative

Test Type: gene mutation test
Test system: mouse lymphoma cells
Metabolic activation: with and without metabolic activation
Method: OECD Test Guideline 476
Result: negative

Genotoxicity in vivo : Test Type: Micronucleus test
Species: Mouse
Method: OECD Test Guideline 474
Result: negative

Carcinogenicity

Not classified based on available information.

Components:**ethyl acetate:**

Result : No evidence of carcinogenicity in animal studies.

Reproductive toxicity

Not classified based on available information.

Components:**ethyl acetate:**

Effects on fertility : Species: Mouse
Application Route: Oral
General Toxicity - Parent: NOAEL: 26,400 mg/kg body weight
Method: OECD Test Guideline 416
Result: Animal testing did not show any effects on fertility.

Effects on foetal development : Test Type: Pre-natal
Species: Rat
Application Route: Inhalation
Developmental Toxicity: NOAEC Parent: 73,300 mg/m³
Method: OECD Test Guideline 414
Result: No teratogenetic, maternal or developmental effects

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STOT - single exposure

May cause drowsiness or dizziness.

Components:**ethyl acetate:**

Exposure routes	:	Inhalation
Target Organs	:	Central nervous system
Assessment	:	May cause drowsiness or dizziness.

STOT - repeated exposure

Not classified based on available information.

Components:**ethyl acetate:**

Exposure routes	:	Inhalation
Remarks	:	No significant adverse effects were reported

Exposure routes	:	Ingestion
Remarks	:	No significant adverse effects were reported

Aspiration toxicity

Not classified based on available information.

Further information**Product:**

Remarks	:	Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting. Concentrations substantially above the TLV value may cause narcotic effects. Solvents may degrease the skin.
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12. ECOLOGICAL INFORMATION**Ecotoxicity****Components:****ethyl acetate:**

Toxicity to fish	:	LC50 (Pimephales promelas (fathead minnow)): 230 mg/l Exposure time: 96 h Method: EPA E03-05
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Toxicity to daphnia and other aquatic invertebrates	:	EC50 (Daphnia magna (Water flea)): 3,090 mg/l Exposure time: 24 h Method: DIN 38412
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NOEC (Daphnia magna (Water flea)): 2.4 mg/l
Exposure time: 21 Days

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Toxicity to algae/aquatic plants : NOEC (Desmodesmus subspicatus (green algae)): > 100 mg/l
Exposure time: 72 h
Method: OECD Test Guideline 201

Toxicity to microorganisms : EC3 (Pseudomonas putida): 650 mg/l
Exposure time: 16 h
Method: DIN 38 412 Part 8

Persistence and degradability**Components:****ethyl acetate:**

Biodegradability : BOD Standard Method
Result: Readily biodegradable.
Biodegradation: 100 %

Bioaccumulative potential**Components:****ethyl acetate:**

Partition coefficient: n-octanol/water : Pow: 0.68
Remarks: measured data

Mobility in soil

No data available

Other adverse effects**Product:**

Additional ecological information : No data available

Components:**ethyl acetate:**

Results of PBT and vPvB assessment : The substance does not meet the criteria for PBT / vPvB according to REACH, Annex XIII

13. DISPOSAL CONSIDERATIONS**Disposal methods**

Waste from residues : Dispose of as hazardous waste in compliance with local and national regulations.
Do not dispose of waste into sewer.
Do not contaminate ponds, waterways or ditches with chemical or used container.
Dispose of contents/ container to an approved waste disposal plant.

Contaminated packaging : Empty remaining contents.
Dispose of as unused product.

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Do not re-use empty containers.
Do not burn, or use a cutting torch on, the empty drum.

14. TRANSPORT INFORMATION**International Regulations****UNRTDG**

UN number : UN 1173
Proper shipping name : ETHYL ACETATE
Class : 3
Packing group : II
Labels : 3

IATA-DGR

UN/ID No. : UN 1173
Proper shipping name : Ethyl acetate
Class : 3
Packing group : II
Labels : Flammable Liquids
Packing instruction (cargo aircraft) : 364
Packing instruction (passenger aircraft) : 353

IMDG-Code

UN number : UN 1173
Proper shipping name : ETHYL ACETATE
Class : 3
Packing group : II
Labels : 3
EmS Code : F-E, S-D
Marine pollutant : no

Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code

Not applicable for product as supplied.

Special precautions for user

The transport classification(s) provided herein are for informational purposes only, and solely based upon the properties of the unpackaged material as it is described within this Safety Data Sheet. Transportation classifications may vary by mode of transportation, package sizes, and variations in regional or country regulations.

15. REGULATORY INFORMATION**Safety, health and environmental regulations/legislation specific for the substance or mixture**

Workplace Safety and Health Act and Workplace Safety and Health (General Provisions) Regulations: This product is subjected to the SDS, labelling, PEL and other requirements in the Act/Regulations.

Environmental Protection and Management Act and : Not applicable
Environmental Protection and Management (Hazardous Substances) Regulations

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Fire Safety (Petroleum and Flammable Materials) : Ethyl acetate
Regulations

16. OTHER INFORMATION

Date format : dd.mm.yyyy

Full text of other abbreviations

ACGIH	: USA. ACGIH Threshold Limit Values (TLV)
SG OEL	: Singapore. Workplace Safety and Health (General Provisions) Regulations - First Schedule Permissible Exposure Limits of Toxic Substances.
ACGIH / TWA	: 8-hour, time-weighted average
SG OEL / PEL (long term)	: Permissible Exposure Level (PEL) Long Term

AIIC - Australian Inventory of Industrial Chemicals; ANTT - National Agency for Transport by Land of Brazil; ASTM - American Society for the Testing of Materials; bw - Body weight; CMR - Carcinogen, Mutagen or Reproductive Toxicant; DIN - Standard of the German Institute for Standardisation; DSL - Domestic Substances List (Canada); ECx - Concentration associated with x% response; ELx - Loading rate associated with x% response; EmS - Emergency Schedule; ENCS - Existing and New Chemical Substances (Japan); ErCx - Concentration associated with x% growth rate response; ERG - Emergency Response Guide; GHS - Globally Harmonized System; GLP - Good Laboratory Practice; IARC - International Agency for Research on Cancer; IATA - International Air Transport Association; IBC - International Code for the Construction and Equipment of Ships carrying Dangerous Chemicals in Bulk; IC50 - Half maximal inhibitory concentration; ICAO - International Civil Aviation Organization; IECSC - Inventory of Existing Chemical Substances in China; IMDG - International Maritime Dangerous Goods; IMO - International Maritime Organization; ISHL - Industrial Safety and Health Law (Japan); ISO - International Organisation for Standardization; KECI - Korea Existing Chemicals Inventory; LC50 - Lethal Concentration to 50 % of a test population; LD50 - Lethal Dose to 50% of a test population (Median Lethal Dose); MARPOL - International Convention for the Prevention of Pollution from Ships; n.o.s. - Not Otherwise Specified; Nch - Chilean Norm; NO(A)EC - No Observed (Adverse) Effect Concentration; NO(A)EL - No Observed (Adverse) Effect Level; NOELR - No Observable Effect Loading Rate; NOM - Official Mexican Norm; NTP - National Toxicology Program; NZIoC - New Zealand Inventory of Chemicals; OECD - Organization for Economic Co-operation and Development; OPPTS - Office of Chemical Safety and Pollution Prevention; PBT - Persistent, Bioaccumulative and Toxic substance; PICCS - Philippines Inventory of Chemicals and Chemical Substances; (Q)SAR - (Quantitative) Structure Activity Relationship; REACH - Regulation (EC) No 1907/2006 of the European Parliament and of the Council concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals; SADT - Self-Accelerating Decomposition Temperature; SDS - Safety Data Sheet; TCSI - Taiwan Chemical Substance Inventory; TDG - Transportation of Dangerous Goods; TECI - Thailand Existing Chemicals Inventory; TSCA - Toxic Substances Control Act (United States); UN - United Nations; UNRTDG - United Nations Recommendations on the Transport of Dangerous Goods; vPvB - Very Persistent and Very Bioaccumulative; WHMIS - Workplace Hazardous Materials Information System

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific

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material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

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